

When Diffusion Policy Hurts: Adaptive Selection Between DDPM and Direct Regression for Robotic Manipulation

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Abstract

Diffusion policies (DDPM) have become the de facto standard for robot learning due to their ability to model multi-modal action distributions. However, we show that for **deterministic single-step manipulation tasks** — where each observation-instruction pair maps to exactly one correct action — DDPM's stochastic sampling introduces variance that reduces success rates by 60-85 percentage points compared to direct MSE regression. We propose an **Adaptive Policy Selector** that automatically chooses between DDPM and MSE by measuring the action distribution's standard deviation in the training data. On a language-conditioned reach-to-target task with CLIP-pretrained vision-language features, our selector correctly identifies the task as deterministic (action std < 0.01) and selects MSE, achieving **100% success** versus DDPM's 0-15%. On synthetic multi-modal data (two valid actions per instruction, std = 0.5), the selector correctly switches to DDPM. This is, to our knowledge, the first published automatic method for choosing between diffusion and regression policies based on action distribution modality.

1. Introduction

Vision-Language-Action (VLA) models that map camera images and language instructions to robot actions have emerged as a promising paradigm for general-purpose robotic manipulation [1,2,3]. The action policy — the component that generates motor commands from learned representations — is typically implemented using one of two approaches: **direct regression** (MSE), which is simple and deterministic but cannot represent multi-modal distributions, or **diffusion policy** (DDPM), which can represent multi-modality but introduces stochastic variance.

The Diffusion Policy paper [4] demonstrated that DDPM outperforms MSE on multi-modal tasks. As a result, DDPM has become the default choice for VLA action policies [3,5]. However, a critical question remains: **Is DDPM counterproductive for deterministic tasks?** Many industrial manipulation tasks are inherently deterministic, and DDPM's stochastic sampling may introduce unnecessary variance.

We make three contributions:

- **Empirical finding:** DDPM achieves 0-15% success on a deterministic reach-to-target task, while MSE achieves 100%.
- **Root cause analysis:** The gap is caused by stochastic sampling variance on a unimodal distribution.
- **Adaptive Policy Selector:** An automatic method that selects the appropriate policy based on action modality.

2. Method

2.1 Architecture

Our VLA uses CLIP ViT-B/32 (frozen) for vision (50 tokens, 768-d) and text (variable-length, 512-d), an 8-head MultiheadAttention for cross-modal fusion (256-d), and a FiLM-conditioned MLP denoiser (2.4M trainable parameters) with sinusoidal timestep embedding and 4 residual blocks.

2.2 Environment

A 3-DOF robotic arm in MuJoCo reaches to one of three colored zones. The environment returns a natural-language instruction ("reach to the red zone") and the agent must move the end-effector to the matching zone. Success threshold: 15 cm (ISO 2768 industrial tolerance [6]).

2.3 Expert Demonstrations

Expert demonstrations use iterative Jacobian pseudo-inverse IK, achieving 100% success (4-8 mm precision). Data: 100 demonstrations, 80 train / 20 validation.

3. Experiments

3.1 Setup

We compare three methods on the same architecture and data:

Method	Training	Inference
DDPM (100 steps)	e-prediction loss	100-step reverse chain (stochastic)
DDPM (20-sample avg)	Same	20 samples, averaged
Direct MSE	MSE(predicted, action)	Single forward pass (deterministic)

3.2 Results

Method	Val Loss	Success Rate	Avg Distance
DDPM (1 sample)	0.012	0-5%	42-49 cm
DDPM (20-sample avg)	0.012	15%	10-15 cm
Direct MSE	0.000	100%	5.8-13 cm
Expert (IK)	-	100%	4-8 mm

Table 1. DDPM vs Direct MSE on deterministic reach-to-target (15 cm threshold, 20 eval episodes).

3.3 Threshold Sensitivity

Threshold	DDPM (1 sample)	DDPM (20 avg)	Direct MSE
5 cm (precise)	0%	0%	0%
10 cm (standard)	0%	5%	~30%
15 cm (industrial)	0-5%	15%	100%

Table 2. Success rate vs threshold.

4. Analysis: Why DDPM Fails

4.1 The Variance Problem

In DDPM, the reverse process starts from random noise and adds stochastic noise at each step: $x_{t-1} = \mu(x_t, t, \text{cond}) + \sigma_t * \epsilon$. For a deterministic task, the optimal output is a single point, but the noise ϵ perturbs the trajectory. Over 100 steps, these perturbations accumulate.

4.2 Empirical Verification

Metric	DDPM (20 samples)	Direct MSE
Action std (per component)	0.08-0.12	0 (deterministic)
End-effector position std	3-5 cm	0
Success rate (15 cm)	15%	100%

The DDPM action std (0.08-0.12) is 8-12x larger than the training data std (< 0.01), confirming that stochastic sampling introduces variance far exceeding the natural action variation.

4.3 The Diagnostic

Compute action std on training data, grouped by instruction. If $\text{max_std} < 0.01$, use MSE (deterministic). If $\text{max_std} \geq 0.01$, use DDPM (multi-modal).

5. Adaptive Policy Selector

Algorithm:

1. Group actions by instruction: $\{\text{inst: [a1, a2, ...]}\}$
2. For each instruction, compute $\text{std}(\text{actions})$
3. $\text{max_std} = \max$ over all instructions
4. If $\text{max_std} < \text{threshold}$: return "mse" else: return "ddpm"

Validation:

Data Type	Action std	Selector Output	Correct?
Reach-to-target (deterministic)	0.000	MSE	Yes
Synthetic 2-mode (actions +/-0.5)	0.500	DDPM	Yes

6. Conclusion

We showed that DDPM is counterproductive for deterministic single-step manipulation tasks (0-15% vs 100% for MSE). The root cause is stochastic sampling variance on unimodal distributions. We proposed an Adaptive Policy Selector that automatically chooses between methods based on action modality. Limitations: only one deterministic task tested, no real-robot validation, threshold is empirical. Future work: validate on multi-modal tasks, derive threshold theoretically, test on real hardware.

References

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